

2023

Refining your trajectory within YellowScan CloudStation.

New POSPac integration.

YellowScan

Designed to innovate.

Summary



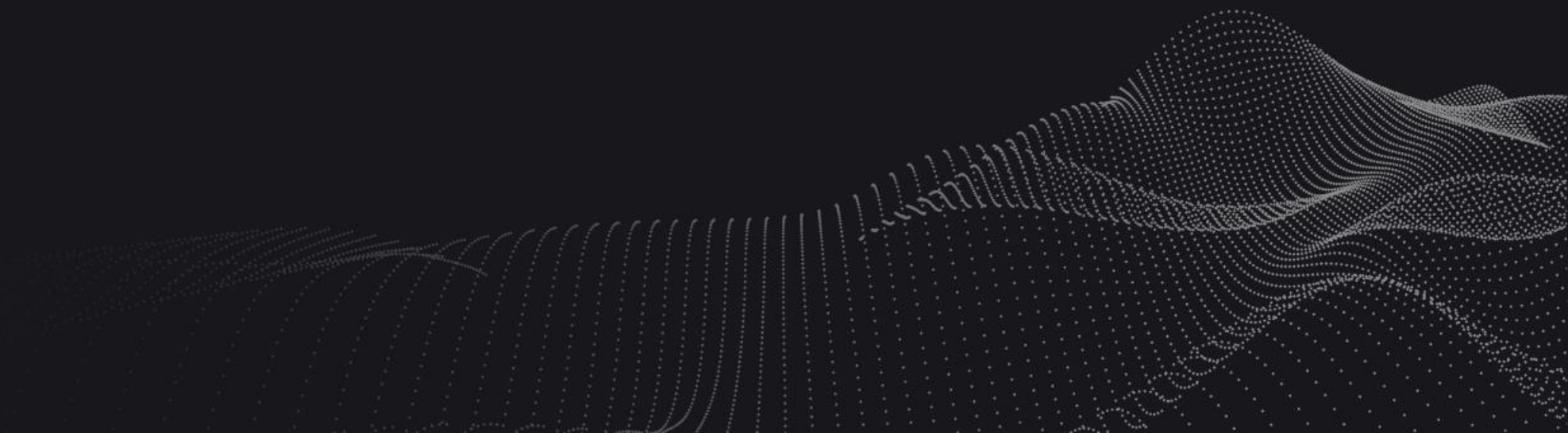
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02 / Overview

03 / Walkthrough

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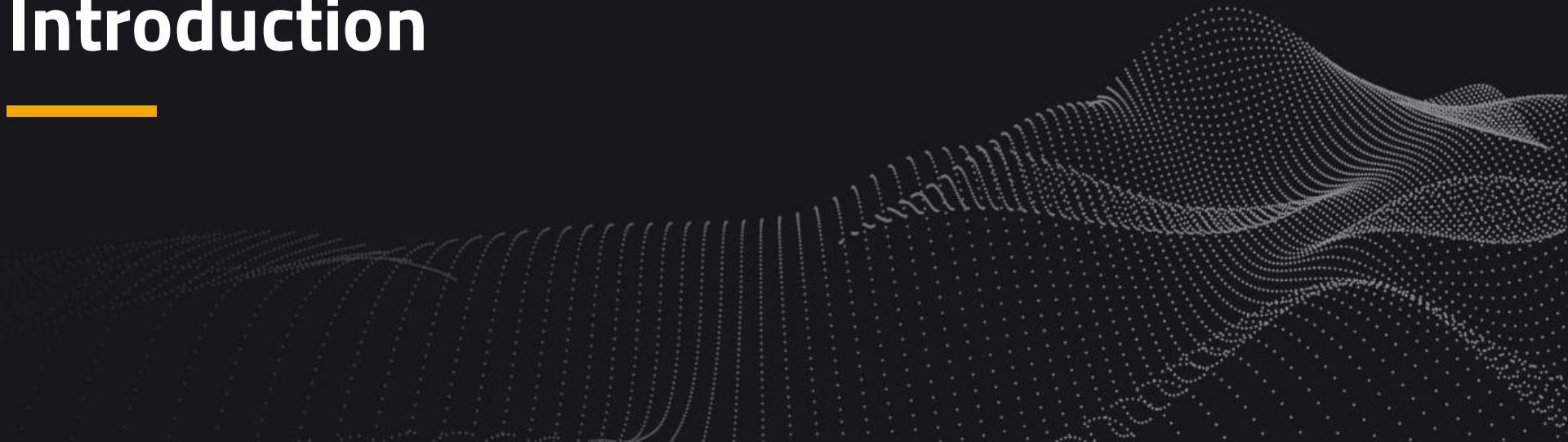
05 / Q&A





01

Introduction



YellowScan CloudStation Software



- Evolutive and user friendly
- Pointcloud generation from raw data
- Visualization and in-depth inspection
- Multiple acquisitions in a single project



Existing features



Strip adjustment

Enhance the accuracy of the pointcloud with a seamless adjustment of flight lines



Terrain

Automatic "ground-non-ground" classification of points



Colorization

Pointcloud colorization from simultaneous LiDAR and camera acquisitions



Export

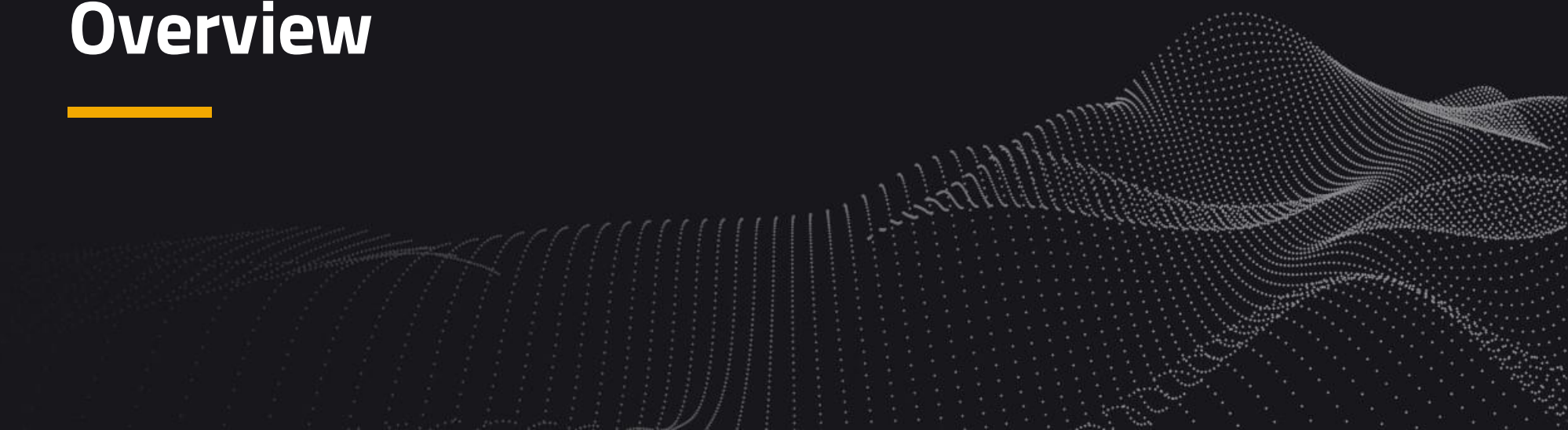
Pointcloud export in LAS/LAZ format

Digital model export: DSM, DTM, DHM



02

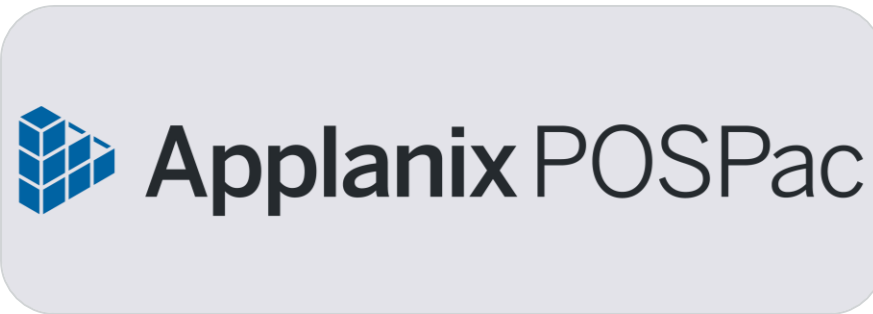
Overview



Current workflow



Data acquisition



Trajectory refinement



Data processing

New workflow



Data acquisition



Trajectory refinement + data processing

Fully integrated within
the YellowScan
CloudStation.

Key benefits

1

/ Reduced complexity

Trajectory refinement in 3 easy steps

2

/ Time saving

Automated one-software solution

3

/ Multiple trajectory processing

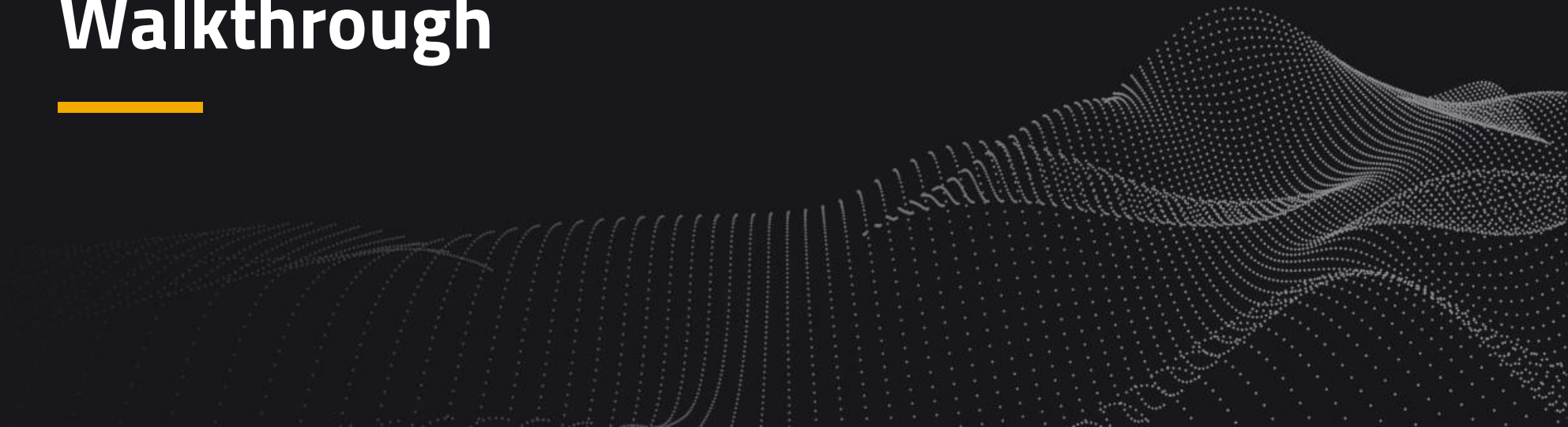
Refine multiple trajectories at once!





03

Walkthrough



Where do I start?

/ Flight selection

/ Same payload configuration

/ Same scanner

Step 1: Payload configuration



PAYLOAD CONFIGURATION

New

Presets

Mapper+ 20221027

Lever arm X

◀ 0.030 m ▶

Lever arm Y

◀ -0.085 m ▶

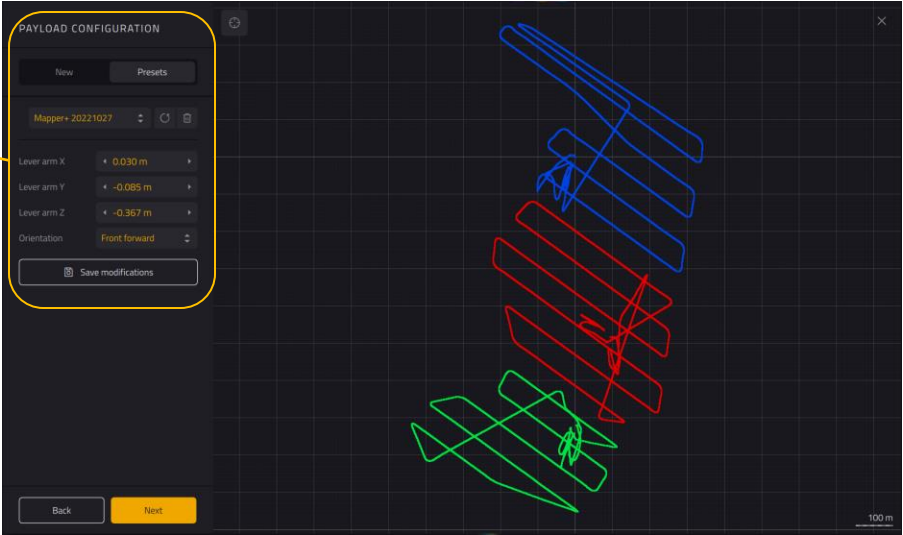
Lever arm Z

◀ -0.367 m ▶

Orientation

Front forward

Save modifications



Step 2: GNSS mode selection



1. RINEX file regrouping

T02 to RINEX conversion

SINGLE BASES

MNTP

mntp117z

mntp117z

mntp117z

mntp117z

2212

22120113y

Back

Next

Importing .t02 files...

2212

4

MNTP

FLIGHTS

YS-20220427-100314

MNTP

YS-20220427-104911

MNTP

YS-20220427-111921

MNTP

Automatic assignment has been applied

3. Automatic base assignment

2. Base station visualization

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Step 3: Export parameters



EXPORT PARAMETERS

Height options ^

Height option **Orthometric** ▾

Geoid model **US2003** ▾

Mapping frame ^

Frame **ITRF2014** ▾

Target epoch **Custom date** ▾

2010-01-01 ✎

EXPORT PARAMETERS

Height options ^

Height option **Orthometric** ▾

Geoid model **US2003** ▾

Mapping frame ^

Frame **ETRF2014** ▾

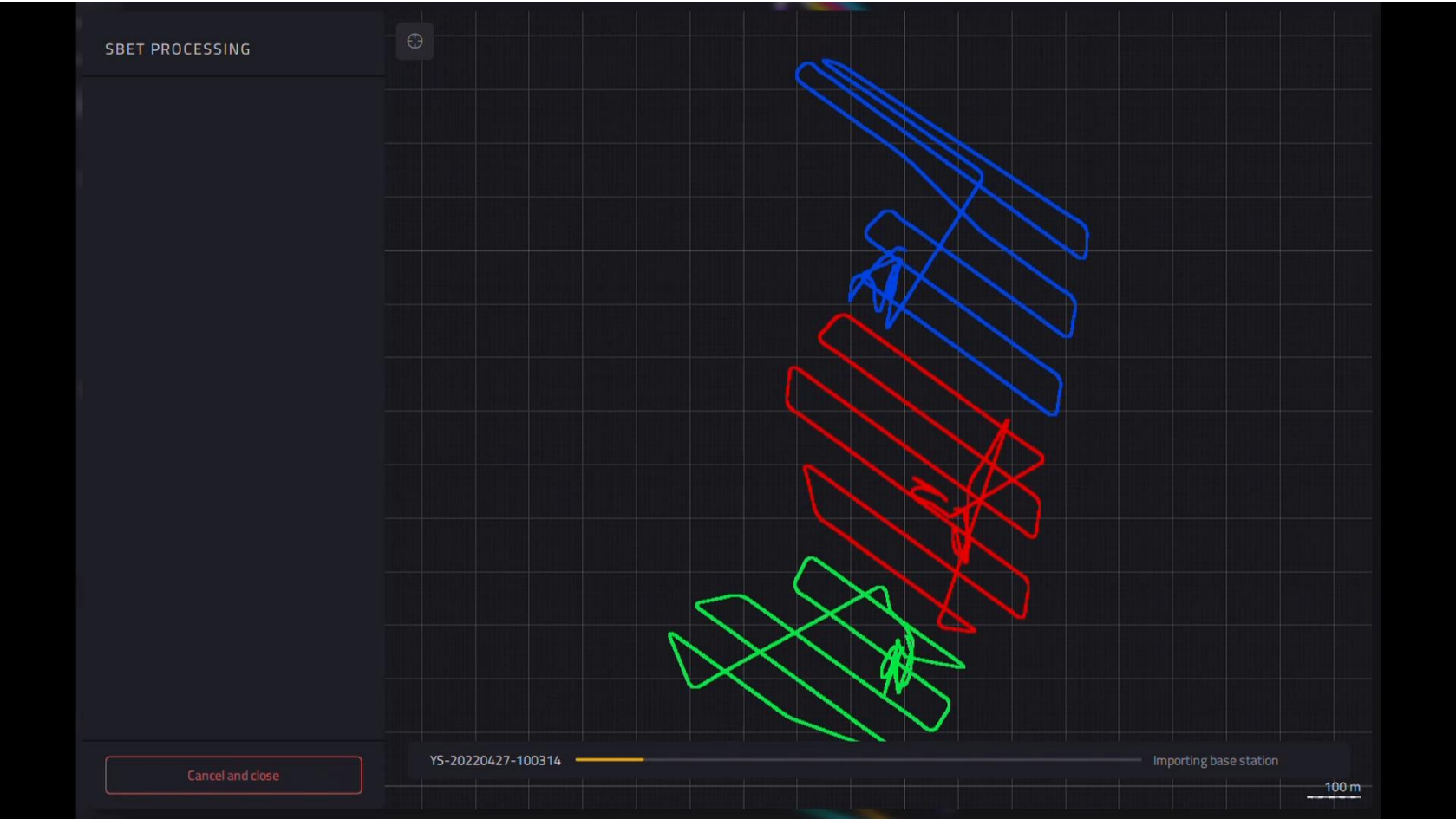
Target epoch **Custom date** ▾

2010-01-01 ✎

Back Process SBET

A 3D point cloud visualization of a building structure, overlaid with colored outlines (red, green, purple) representing different parts of the model. A scale bar indicates 100 m.

Processing



Processing



SBET PROCESSING

☒

YS-20220427-100314

☒

☒ POSDataImport

☒ BaseStationImport

☒ GNSSInertial

☒ Export

☒ QcReport

☒

YS-20220427-104911

☒

☒

YS-20220427-111921

☒

SBET PROCESSING

☒ YS-20220427-100314 ☒

☒ POSDataImport

☒ BaseStationImport

☒ GNSSInertial

☒ Export

☒ QcReport

☒ YS-20220427-104911 ☒

☒ YS-20220427-111921 ☒

Close

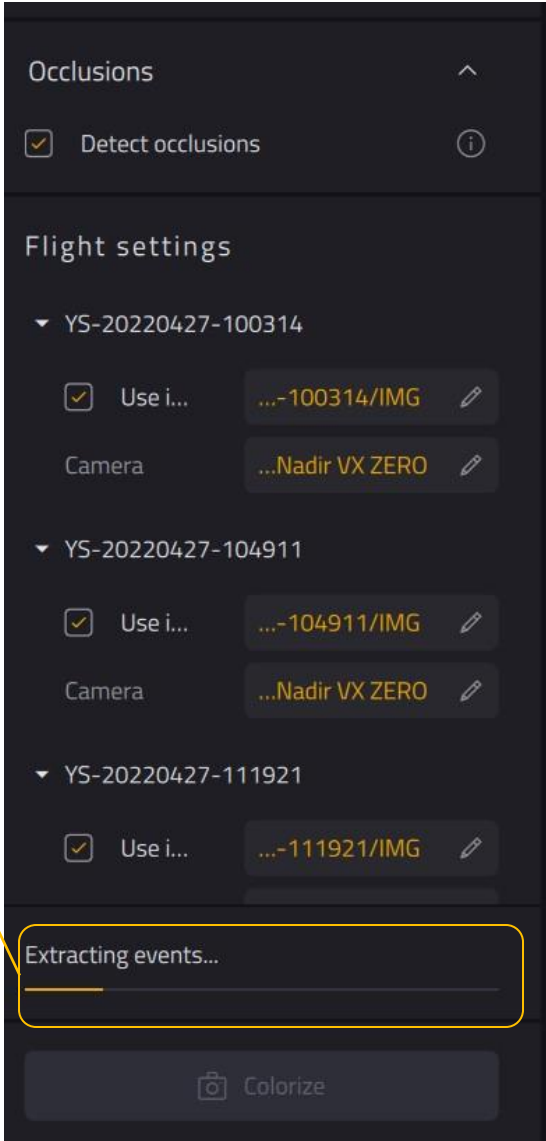
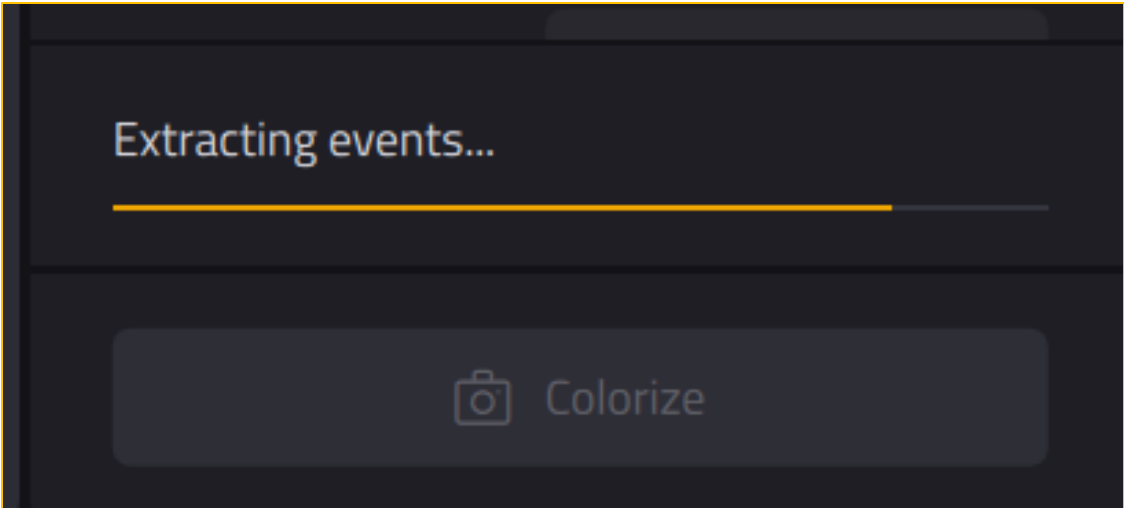
Apply SBET

Close

Apply SBET

A 3D point cloud visualization of a building structure. The building is outlined with blue, red, and green lines, indicating different processing stages or areas. A scale bar at the bottom right indicates 100 m.

Camera event extraction





04

Technical specifications

Technical specifications

1

/ Licensing

Included in the Core license of the YellowScan CloudStation

2

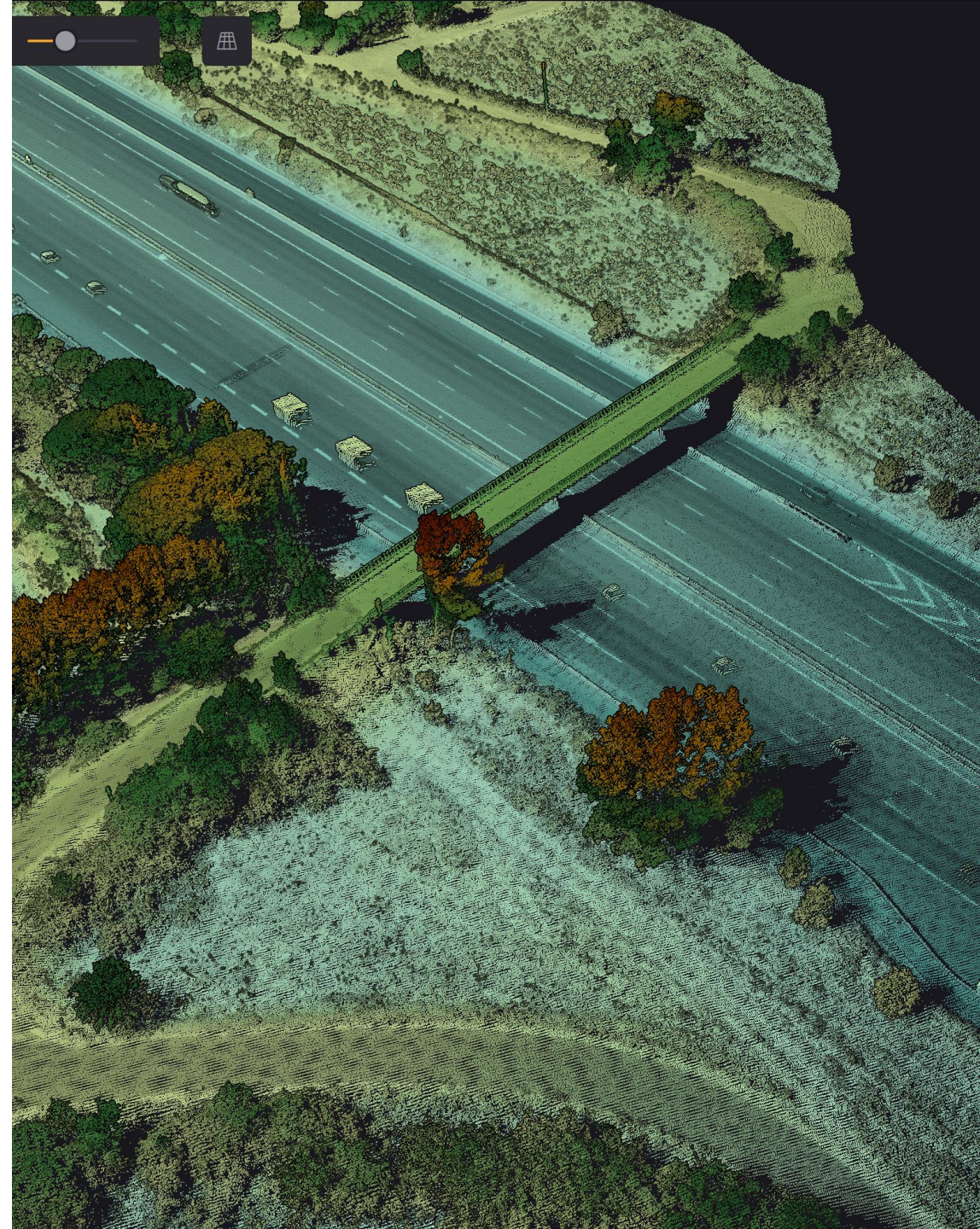
/ POSPac version

Works with POSPac MMS and UAV (8.5 or newer)

3

/ Availability

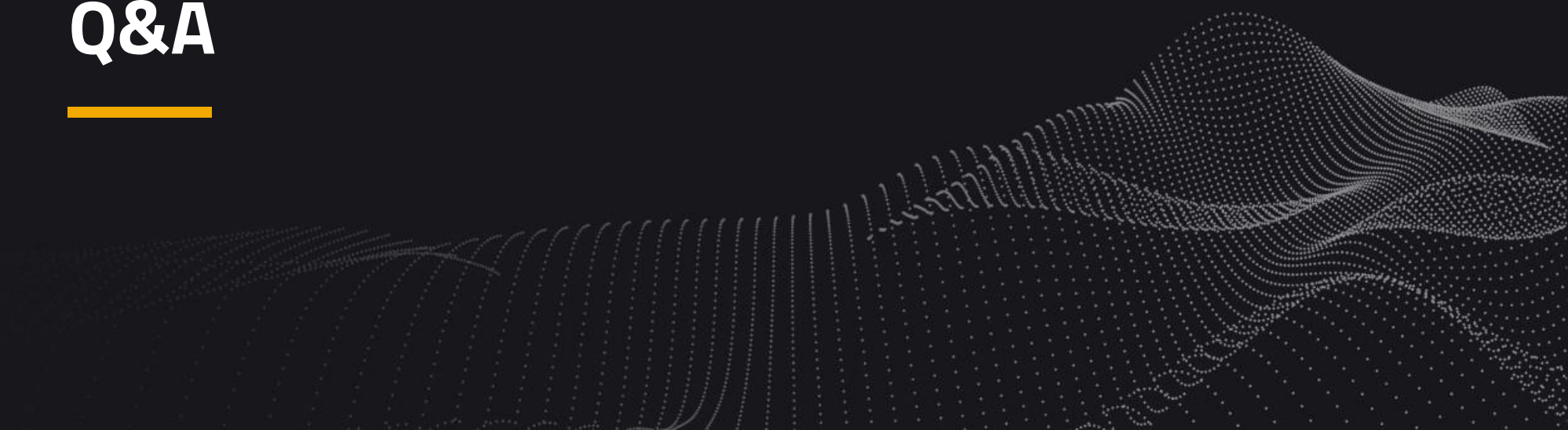
The trajectory refinement integration is available in the last version of the CloudStation that came out May 17th, 2023

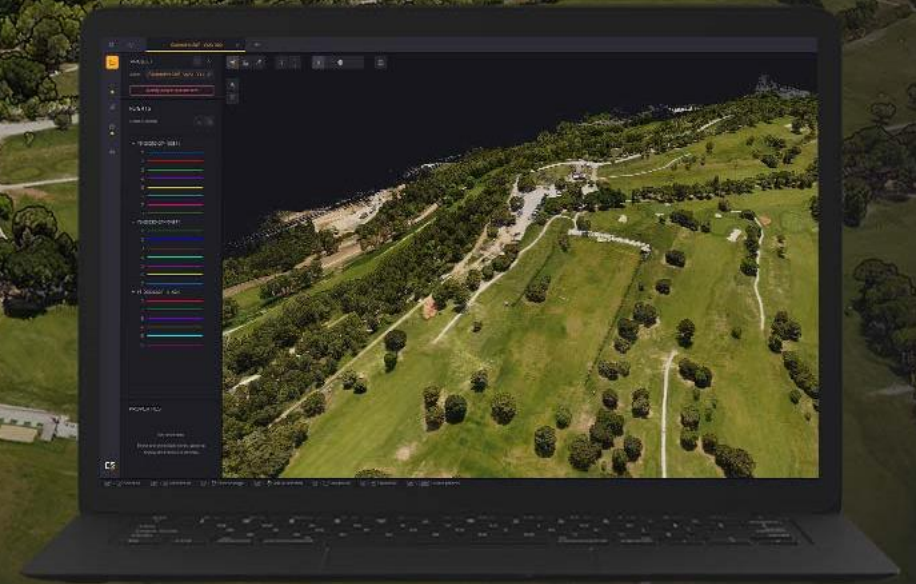




05

Q&A





Thank you for your attention!
